Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A spherically symmetric charge distribution is characterised by a charge density having the following variations:
$\rho(\mathrm{r})=\rho_{\mathrm{o}}\left(1-\frac{\mathrm{r}}{\mathrm{R}}\right)$ for $\mathrm{r}<\mathrm{R}$ $\rho(\mathrm{r})=0$ for $\mathrm{r} \geq \mathrm{R}$
Where r is the distance from the centre of the charge distribution ρ_0 is a constant. The electric field at an internal point $(\mathbf{r}<\mathrm{R})$ is:

(A) $\frac{\rho_{\mathrm{o}}}{4 \varepsilon_{\mathrm{o}}}\left(\frac{\mathrm{r}}{3}-\frac{\mathrm{r}^2}{4 \mathrm{R}}\right)$

(B) $\frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3}-\frac{r^2}{4 R}\right)$

(C) $\frac{\rho_{\mathrm{o}}}{3 \varepsilon_{\mathrm{o}}}\left(\frac{\mathrm{r}}{3}-\frac{\mathrm{r}^2}{4 \mathrm{R}}\right)$

(D) $\frac{\rho_{\mathrm{o}}}{12 \varepsilon_{\mathrm{o}}}\left(\frac{\mathrm{r}}{3}-\frac{\mathrm{r}^2}{4 \mathrm{R}}\right)$

Q.2

A semi-circular arc of radius ' a ' is charged uniformly and the charge per unit length is λ. The electric field at the centre of this arc is

(A) $\frac{\lambda}{2 \pi \varepsilon_0 a}$

(B) $\frac{\lambda}{2 \pi \varepsilon_0 a^2}$

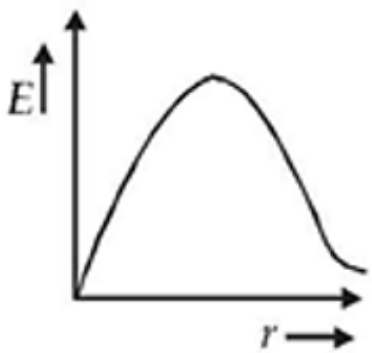
(C) $\frac{\lambda}{4 \pi^2 \varepsilon_0 a}$

(D) $\frac{\lambda^2}{2 \pi \varepsilon_0 a}$

Q.3

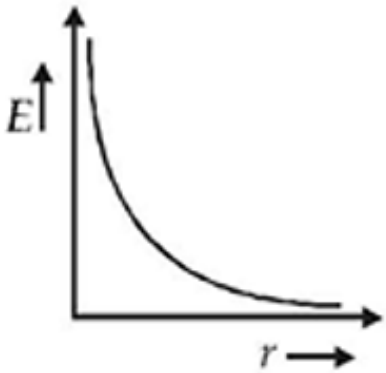
In a uniformly charged sphere of total charge Q and radius R, the electric field E is plotted as function of distance from the centre. The graph which would correspond to the above will be:

(A)

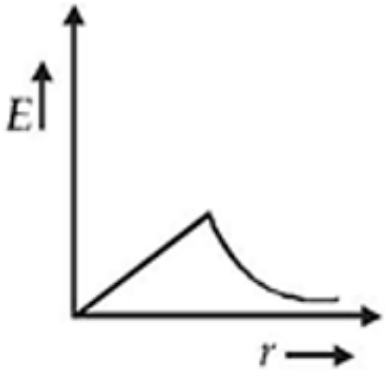


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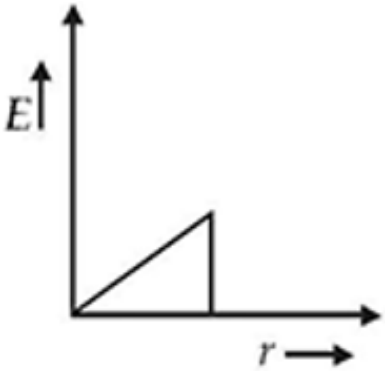
(B)



(C)

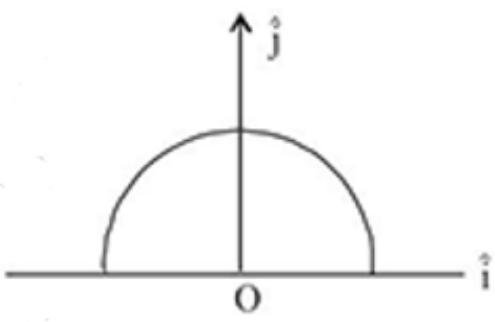


(D)



Q.4

A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net electric field $\vec{E}$ at the centre O is



(A) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

(B) $-\frac{q}{4\pi^2\varepsilon_n r^2}\hat{j}$

(C) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

(D) $\frac{q}{2p^2e_0 r^2}\hat{j}$

Q.5

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A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = \kappa r^a$, where κ and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, find the value of a

(A) 1

(B) 2

(C) 3

(D) 4

Q.6

An uniform electric field E exists along positive x-axis. The work done in moving a charge $0.5\mathrm{C}$ through a distance $2\ \mathrm{m}$ along a direction making an angle 60° with x -axis is $10\ \mathrm{J}$. Then what is the magnitude of electric field (in Vm^{-1})?

(A) 15

(B) 20

(C) 25

(D) 30

Q.7

The surface charge density of a thin charged disc of radius R is σ. The value of the electric field at the centre of the disc
is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc reduces (in percent) by

Answer should be rounded off to two decimal places.

(A) 714.3

(B) 0.7143

(C) 7.143

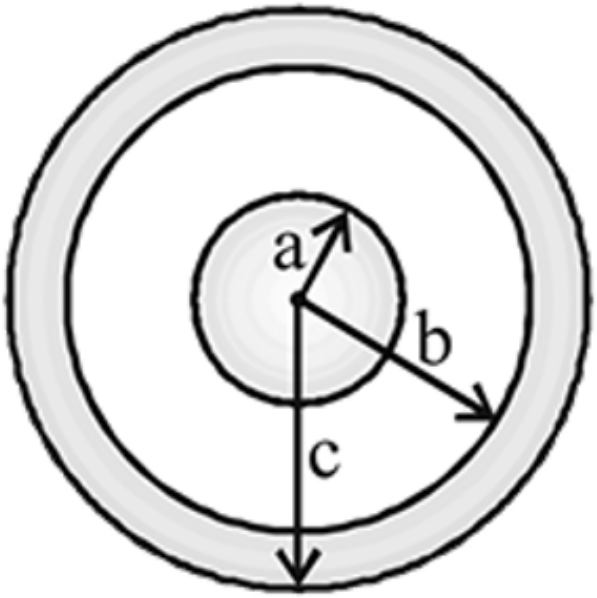
(D) 71.43

Q.8

A solid conducting sphere of radius a has a net positive charge $2Q$. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge $-$ Q.The surface charge density

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on the inner and outer surfaces of the spherical shell will be



(A) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(B) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(C) $0, \frac{Q}{4\pi c^2}$

(D) None of the above

Q.9

A uniformly charged conducting sphere of $4.4\ \mathrm{m}$ diameter has a surface charge density of 60 $\mu C\ \mathrm{m}^{-2}$. The charge on the sphere is

(A) $7.3 \times 10^{-3} C$

(B) $3.7 \times 10^{-6} \mathrm{C}$

(C) $7.3 \times 10^{-6} C$

(D) $3.7 \times 10^{-3} C$

Q.10

A rod of length $2.4\ \mathrm{m}$ and radius $4.6\ \mathrm{mm}$ carries a negative charge of $4.2 \times 10^{-7} \mathrm{C}$ spread uniformly over it surface. The electric field near the mid-point of the rod, at a point on its surface is

(A) $-8.6 \times 10^5\ \mathrm{NC}^{-1}$

(B) $8.6 \times 10^4\ \mathrm{NC}^{-1}$

(C) $-6.7 \times 10^5\ \mathrm{NC}^{-1}$

(D) $6.7 \times 10^4\ \mathrm{NC}^{-1}$

Q.11

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If electric field in a region is radially outward with magnitude $\mathrm{E}=\mathrm{Ar}$, the charge contained in a sphere of radius r centred at the origin is

(A) $\frac{1}{4\pi\varepsilon_{\mathrm{o}}}\mathrm{Ar}^3$

(B) $4\pi\varepsilon_0\mathrm{Ar}^3$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{A}{r^3}$

(D) $\frac{4\pi\varepsilon_0\,\mathrm{A}}{\mathrm{r}^3}$

Q.12

The electric field intensity just sufficient to balance the earth's gravitational attraction on an electron will be: (given mass and charge of an electron respectively are $9.1\times10^{-31}\ \mathrm{kg}$ and $1.6\times10^{-19}\mathrm{C}.$)

(A) $-5.6\times10^{-11}\ \mathrm{N/C}$

(B) $-4.8\times10^{-15}\ \mathrm{N/C}$

(C) $-1.6\times10^{-19}\ \mathrm{N/C}$

(D) $-3.2\times10^{-19}\ \mathrm{N/C}$

Q.13

The insulation property of air breaks down when the electric field is $3\times10^6\mathrm{Vm}^{-1}$. The maximum charge that can be given to a sphere of diameter $5\ \mathrm{m}$ is approximately

(A) $2\times10^{-2}\mathrm{C}$

(B) $2\times10^{-3}\mathrm{C}$

(C) $2\times10^{-4}\mathrm{C}$

(D) $2\times10^{-5}\mathrm{C}$

Q.14

A hollow insulated conduction sphere is given a positive charge of $10\mu\mathrm{C}$. What will be the electric field at the centre of the sphere if its radius is $2\ \mathrm{m}$?

(A) Zero

(B) $5\mu\mathrm{Cm}^{-2}$

(C) $20\mu\mathrm{Cm}^{-2}$

(D) $8\mu\mathrm{Cm}^{-2}$

Q.15

The number of electric lines of force that radiate outwards from one coulomb of charge in vacuum is

(A) 1.13×10^{11}

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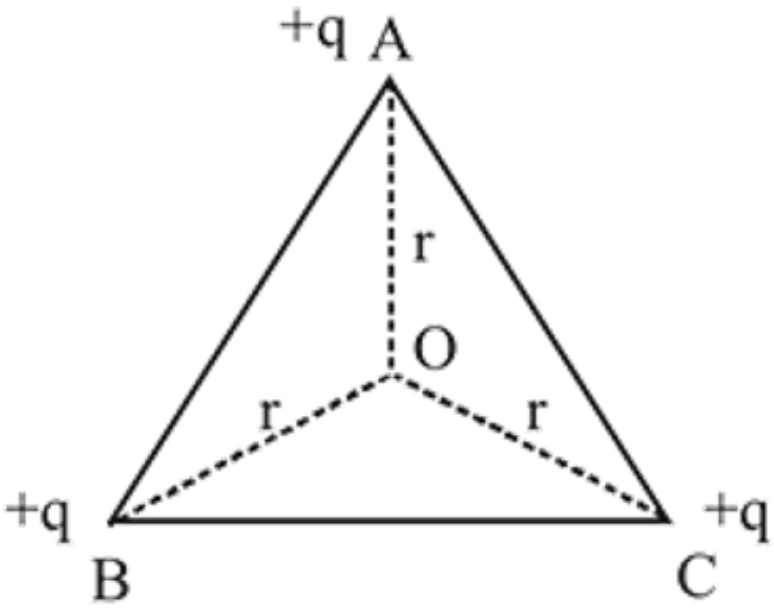
(B) 1.13×10^{10}

(C) 0.61×10^{11}

(D) 0.61×10^{9}

Q.16

ABC is an equilateral triangle. Charges $+\mathrm{q}$ are placed at each corner as shown in fig. The electric intensity at centre O will be



(A) $\frac{1}{4\pi\in_0}\frac{\mathrm{q}}{\mathrm{r}}$

(B) $\frac{1}{4\pi\in_0}\frac{\mathrm{q}}{\mathrm{r}^2}$

(C) $\frac{1}{4\pi\in_0}\frac{3q}{r^2}$

(D) zero

Q.17

Let $\rho(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point ' P ' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is :

(A) $\frac{Q}{4\pi\in_0 r_1^2}$

(B) $\frac{Qr_1^2}{4\pi\in_0 R^4}$

(C) $\frac{Qr_1^2}{3\pi\in_0 R^4}$

(D) 0

Q.18

In the figure the electric lines on the right have twice the separation of those on the left. If a charge particle takes time t to move a distance x in left region, then it will take time to travel the same distance in the right

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side region is :



(A) $\frac{t}{2}$

(B) t

(C) $\sqrt{2}t$

(D) $2t$

Q.19

n identical point charges are kept symmetricall on the periphery of the circle $x^2 + y^2 = R^2$ in xy plane. The resultant electric field at $(0, 0, \mathrm{R})$ is E_1
and at $(0, 0, 2\mathrm{R})$ is E_2. The ratio of $\frac{\mathrm{E}_1}{\mathrm{E}_2}$ is

(A) $\frac{5\sqrt{5}}{4\sqrt{2}}$

(B) $\frac{5}{2}$

(C) $\frac{5}{4}$

(D) $\frac{5\sqrt{5}}{2\sqrt{2}}$

Q.20

Given point charges $+8\mu C$, $-1\mu C$, $-1\mu C$ and $+8\mu C$ are fixed at the points-$\sqrt{\frac{27}{2}}$ m,-$\sqrt{\frac{3}{2}}$ m,+$\sqrt{\frac{3}{2}}$ m and $+\sqrt{\frac{27}{2}}$ m respectively on the y-axis, A particle of mass 6×10^{-4}kg and charge $+0.1\mu C$ moves along the x - $axis$. If Its speed at $x = +\infty$ is v_0. Then find the minimum value of v_0 in m s^{-1} for which the particle will cross the origin. $\left(\frac{1}{4}\ \pi\varepsilon_0 = 9 \times 10^9\ \mathrm{Nm^2\ C^{-2}}\right)$.

(A) 1

(B) 2

(C) 3

(D) 4

Answers & Solutions

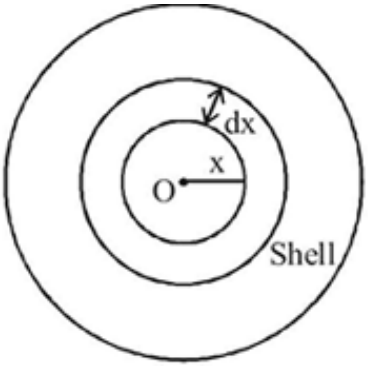
Q.1 Answer:

$\frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$

Solution:

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Let us consider a spherical shell of radius x and thickness dx.



Charge on this shell

$\mathrm{dq} = \rho.4\pi \mathrm{x}^2 \mathrm{dx} = \rho_0 \left(1 - \frac{\mathrm{x}}{\mathrm{R}}\right) \cdot 4\pi \mathrm{x}^2 \, \mathrm{dx}$

∴ Total charge in the spherical region from centre to $r(r < R)$ is

$\mathrm{q} = \int \mathrm{dq} = 4\pi\rho_0 \int_0^{\mathrm{r}} \left(1 - \frac{\mathrm{x}}{\mathrm{R}}\right) \mathrm{x}^2 \, \mathrm{dx}$

$= 4\pi\rho_0 \left[\frac{x^3}{3} - \frac{x^4}{4R}\right]_0^r$

$= 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4R}\right]$

$= 4\pi\rho_0 r^3 \left[\frac{1}{3} - \frac{r}{4R}\right]$

∴ Electric field at $\mathrm{r}, \mathrm{E} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\mathrm{q}}{\mathrm{r}^2}$

$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{4\pi\rho_0 r^3}{r^2} \left[\frac{1}{3} - \frac{r}{4R}\right]$

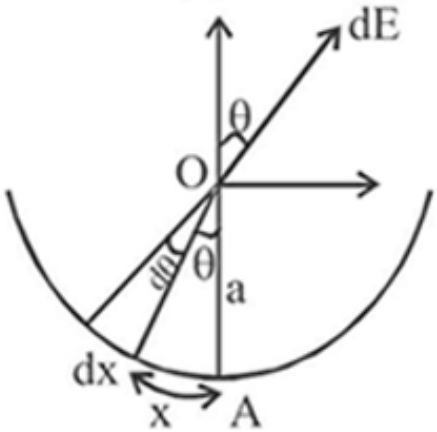
$= \frac{\rho_0}{\varepsilon_0} \left[\frac{r}{3} - \frac{r^2}{4R}\right]$

Q.2 Answer:

$\frac{\lambda}{2\pi\varepsilon_0 a}$

Solution:

λ = linear charge density; Charge on elementary portion $dx = \lambda dx$.



Electric field at $O, dE = \frac{\lambda dx}{4\pi\varepsilon_0 a^2}$

Horizontal electric field, i.e., perpendicular to AO, will be cancelled. Hence, net electric field = addition of all electrical fields in direction of $AO = \Sigma dE \cos\theta$

$\Rightarrow E = \int \frac{\lambda dx}{4\pi\varepsilon_0 a^2} \cos\theta$

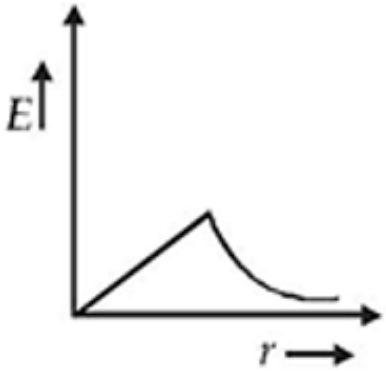
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Also, $d\theta = \frac{dx}{a}$ or $dx = ad\theta$

$E = \int_{-\pi/2}^{\pi/2} \frac{\lambda \cos\theta d\theta}{4\pi\varepsilon_0 a} = \frac{\lambda}{4\pi\varepsilon_0 a}[\sin\theta]_{-\pi/2}^{\pi/2}$

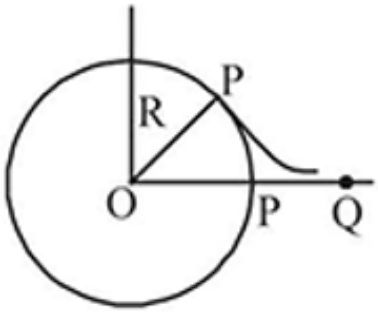
$= \frac{\lambda}{4\pi\varepsilon_0 a}[1-(-1)] = \frac{\lambda}{2\pi\varepsilon_0 a}$

Q.3 Answer:



Solution:

$\vec{E}$ **inside the charged sphere**



$\vec{E}_{in} = 0$ **...(i)**

$\vec{E}$ **on the surface of the charged sphere** $\vec{E}_s = \frac{1}{4\pi\in_0}\frac{q}{R^2}$ **i.e.,** $\vec{E}_s \propto \frac{1}{R^2}\hat{n}$ **... (ii)**

$\vec{E}$ **on any point away from the uniformly charged sphere is given** $\vec{E} = \frac{1}{4\pi\in_0}\frac{q}{r^2}\hat{n}$

$\vec{E} \propto \frac{1}{r^2}\hat{n}$

$\because R$ **is the radius of the sphere, which is constant,**

thus $\vec{E}$ **is maximum and constant at the surface of the sphere. But decreases on moving away from the surface of the uniformally charged sphere.**

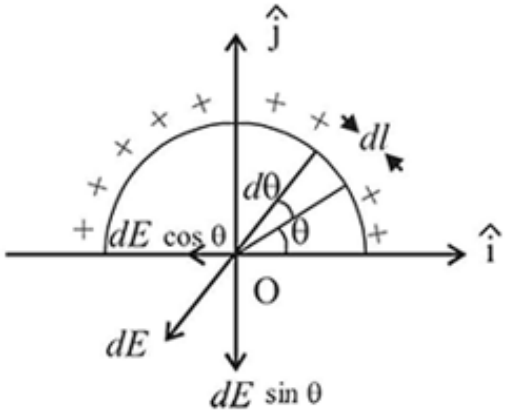
Q.4 Answer:

$-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

Solution:

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Let us consider a differential element dl. charge on this element.



$$\begin{aligned} dq &= \left(\frac{q}{\pi r}\right) dl \\ &= \frac{q}{\pi r}(r d\theta) \quad (\because dl = r d\theta) \\ &= \left(\frac{q}{\pi}\right) d\theta \end{aligned}$$

Electric field at O due to dq is

$$dE = \frac{1}{4\pi \in_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi \in_0} \cdot \frac{q}{\pi r^2} d\theta$$

The component $dE\cos\theta$ will be counter balanced by another element on left portion. Hence resultant field at O is the resultant of the component $dE\sin\theta$ only.

$$\begin{aligned} \therefore E &= \int dE \sin\theta = \int_0^{\pi} \frac{q}{4\pi^2 r^2 \in_0} \sin\theta d\theta \\ &= \frac{q}{4\pi^2 r^2 \in_0} [-\cos\theta]_0^{\pi} \\ &= \frac{q}{4\pi^2 r^2 \in_0} (+1+1) = \frac{q}{2\pi^2 r^2 \in_0} \end{aligned}$$

The direction of E is towards negative y-axis. $\therefore \vec{E} = -\frac{q}{2\pi^2 r^2 \in_0}\hat{j}$

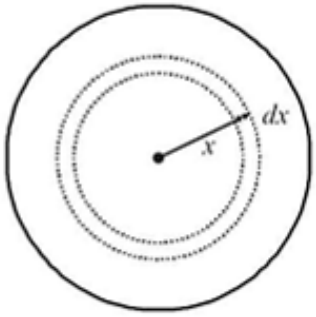
Q.5 Answer:

2

Solution:

Let us consider a spherical shell of radius x and thickness dx. The volume of this shell is $4\pi x^2(dx)$. The charge enclosed in this spherical shell is

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$dq = \left(4\pi x^2\right) dx \times kx^a$

$\therefore \quad dq = 4\pi k x^{2+a} dx$

For $r = R$:

The total charge enclosed in the sphere of radius R is

$$Q = \int_0^R 4\pi k x^{2+a} dx = 4\pi k \frac{R^{3+a}}{3+a} .$$

$\therefore$ **The electric field at $r = R$ is**

$$E_1 = \frac{1}{4\pi\varepsilon_0} \frac{4\pi k R^{3+a}}{(3+a)R^2} = \frac{1}{4\pi\varepsilon_0} \frac{4\pi k}{3+a} R^{1+a}$$

For $r = R/2$:

The total charge enclosed in the sphere of radius $R/2$ is

$$Q' = \int_0^{R/2} 4\pi k x^{2+a} dx = \frac{4\pi k (R/2)^{3+a}}{3+a}$$

$\therefore$ **The electric field at $r = R/2$ is**

$E_2 = \frac{1}{4\pi\varepsilon_0} \frac{4\pi k}{3+a} \frac{(R/2)^{3+a}}{(R/2)^2} = \frac{1}{4\pi\varepsilon_0} \frac{4\pi k}{3+a} \left(\frac{R}{2}\right)^{1+a}$

Given, $E_2 = \frac{1}{8} E_1$

$$\therefore \quad \frac{1}{4\pi\varepsilon_0} \frac{4\pi k}{(3+a)} \left(\frac{R}{2}\right)^{1+a} = \frac{1}{2^3} \times \frac{1}{4\pi\varepsilon_0} \frac{4\pi k}{3+a} R^{1+a}$$

$$\Rightarrow \quad 1 + a = 3 \Rightarrow a = 2$$

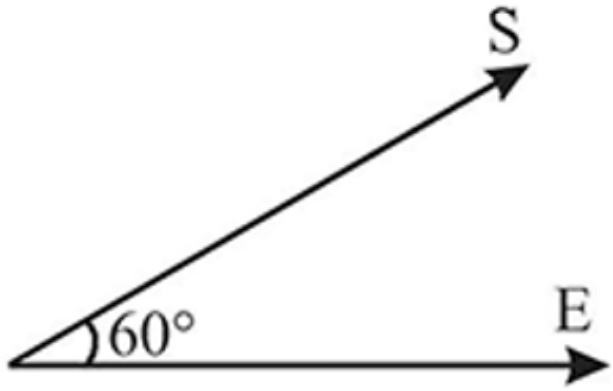
Q.6 Answer:

20

Solution:

Force acting on the charged particle due to electric field $= q\vec{E}$

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work done in moving through distance S,

$$\mathrm{W} = \mathrm{q}\overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{S}} = (\mathrm{qE}) \times \mathrm{S} \times \cos\theta$$

$$\therefore 10\ \mathrm{J} = (0.5\mathrm{C}) \times \mathrm{E} \times 2\cos 60^\circ$$

$$\mathrm{E} = 10 \times 2 = 20\mathrm{NC}^{-1} = 20\mathrm{Vm}^{-1}$$

Q.7 Answer:

71.43

Solution:

Electric field intensity at the centre of the disc. $\mathrm{E} = \frac{\sigma}{2\epsilon_0}$ **(given)**

Electric field along the axis at any distance x **from the centre of the disc**

$$\mathrm{E}' = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{\mathrm{x}}{\sqrt{\mathrm{x}^2 + \mathrm{R}^2}}\right)$$

From question, $\mathrm{x} = \mathrm{R}$ **(radius of disc)**

$$\therefore \mathrm{E}' = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{\mathrm{R}}{\sqrt{\mathrm{R}^2 + \mathrm{R}^2}}\right)$$

$$= \frac{\sigma}{2\epsilon_0}\left(\frac{\sqrt{2}\mathrm{R} - \mathrm{R}}{\sqrt{2}\mathrm{R}}\right) = \frac{4}{14}\mathrm{E}$$

$\therefore \%$ **reduction in the value of electric field**

$$= \frac{\left(\mathrm{E} - \frac{4}{14}\mathrm{E}\right) \times 100}{\mathrm{E}} = \frac{1000}{14}\% \simeq 71.428\%$$

Q.8 Answer:

$-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

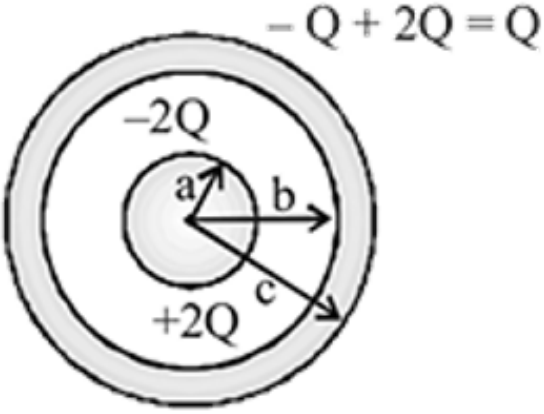
Solution:

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Surface charge density $(\sigma) = \frac{\text{Charge}}{\text{Surface area}}$

So $\sigma_{\text{inner}} = \frac{-2Q}{4\pi b^2}$

and $\sigma_{\text{Outer}} = \frac{Q}{4\pi c^2}$



Q.9 Answer:

$3.7 \times 10^{-3} C$

Solution:

Here, $D = 2r = 4.4$ m, **or** $= r = 2.2$ m

$\sigma = 60 \mu C m^{-2}$

Charge on the sphere, $q = \sigma \times 4\pi r^2 = 60 \times 10^{-6} \times 4 \times \frac{22}{7} \times (2.2)^2 = 3.7 \times 10^{-3}$C

Q.10 Answer:

-6.7×10^5 NC^{-1}

Solution:

Here, $\ell = 2.4$ m, $r = 4.6$ mm $= 4.6 \times 10^{-3}$ m

Linear charge density, $\lambda = \frac{q}{\ell} = \frac{-4.2 \times 10^{-7}}{2.4} = -1.75 \times 10^{-7}$Cm^{-1}

Electric field, $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

$= \frac{-1.75 \times 10^{-7}}{2 \times 3.14 \times 8.854 \times 10^{-12} \times 4.6 \times 10^{-3}}$

$= -6.7 \times 10^5$ NC^{-1}

Q.11 Answer:

$4\pi\varepsilon_0 \text{Ar}^3$

Solution:

$\text{E} = \frac{\text{q}}{4\pi\varepsilon_\text{o}\text{r}^2} \Rightarrow \text{Ar} = \frac{\text{q}}{4\pi\in_0\text{r}^2}$

$\Rightarrow \text{q} = 4\pi\varepsilon_\text{o}\text{Ar}^3$

Q.12 Answer:

-5.6×10^{-11} N/C

Solution:

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$-eE = mg$

$\overrightarrow{E} = -\frac{9.1\times10^{-31}\times10}{1.6\times10^{-19}} = -5.6\times10^{-11}\ N/C$

Q.13 Answer:

$2\times10^{-3}C$

Solution:

$E = \frac{kQ}{r^2} \Rightarrow Q = \frac{E\times r^2}{k} = \frac{3\times10^6\times(2.5)^2}{9\times10^9}$

$= 2\times10^{-3}C$

Q.14 Answer:

Zero

Solution:

Electric field at centre of narrow sphere is always zero..

Q.15 Answer:

1.13×10^{11}

Solution:

Here, $q = 1C, \varepsilon_0 = 8.85\times10^{-12}C^2N^{-1}\ m^{-2}$

Number of lines of force = Electric force $= \frac{q}{\varepsilon_0} = \frac{1}{8.85\times10^{-12}} = 1.13\times10^{11}$

Q.16 Answer:

zero

Solution:

Unit positive charge at O **will be repelled equally by three charges at the three corners of triangle.**

By symmetry, resultant E **at** O **would be zero.**

Q.17 Answer:

$\frac{Qr_1^2}{4\pi\in_0 R^4}$

Solution:

Consider a differential thickness dr at a radius r**.**

We get the area for this differential thickness as $dA = 4\pi r^2 dr$**,**

Thus we get the electric field at this point as $dE = \frac{kdQ}{r_1^2}$

or

$dE = \frac{1}{4\pi\varepsilon_0}\frac{Qr4\pi r^2 dr}{\pi R^4 r_1^2}$

$E = \frac{Q}{4\pi\epsilon_0 r_1^2\pi R^4}\int_{r=0}^{r_1} 4\pi r^3 dr$

$= \frac{Qr_1^2}{4\pi\epsilon_0 R^4}$

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Q.18 Answer:

$\sqrt{2}t$

Solution:

$x = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{Eq}{m}\right)t^2$ **and** $x = \frac{1}{2}\left(\frac{Eq}{2m}\right)t'^2$

$\therefore t' = \sqrt{2}t$

Q.19 Answer:

$\frac{5\sqrt{5}}{4\sqrt{2}}$

Solution:

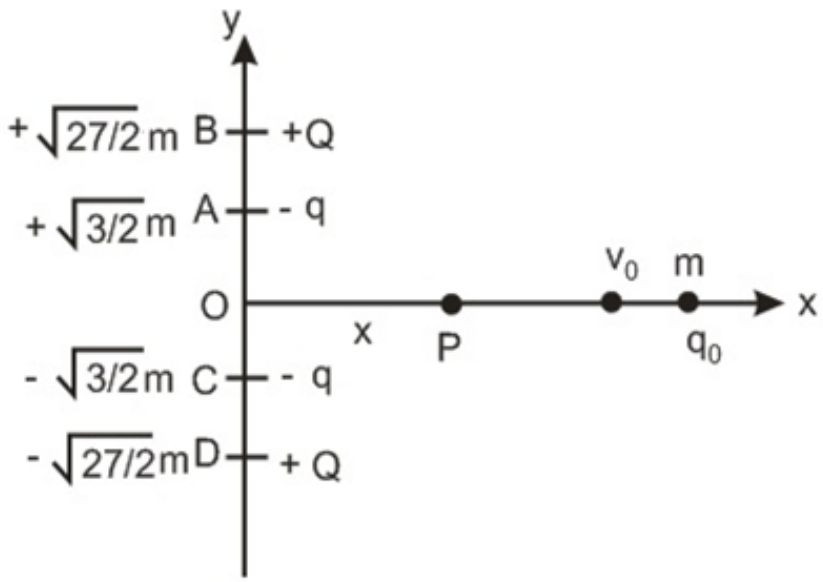
$E_1 = \frac{nkQR}{(R^2+R^2)^{3/2}}, E_2 = \frac{nkQ(2R)}{(R^2+(2R)^2)^{3/2}}$

$\therefore = \frac{E_1}{E_2} = \frac{5\sqrt{5}}{4\sqrt{2}}$

Q.20 Answer:

3

Solution:



In the figure q = 1μC = 10^{-6}C, q_0 = + 0. 1μC = 10^{-7}C

and m = 6x 10^{-4} kg and Q = 8μC = 8 x 10^{-6} C

Let P be any point at a distance x from origin O. Then

$AP = CP = \sqrt{\frac{3}{2} + x^2}$

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$BP = DP = \sqrt{\frac{27}{2} + x^2}$

Electric potential at point P will be

$V = \frac{2KQ}{BP} - \frac{2Kq}{AP}$

where $K = \frac{1}{4\pi\varepsilon_0}$

= 9 x 10^9 Nm^2/C^2

Therefore V = 2 x 9 x 10^9 $\frac{8 \times 10^{-6}}{\sqrt{\frac{27}{2} + x^2}} - \frac{10^{-6}}{\sqrt{\frac{3}{2} + x^2}}$

V = 1. 8 x 10^4 $\frac{8}{\sqrt{\frac{27}{2} + x^2}} - \frac{10^{-6}}{\sqrt{\frac{3}{2} + x^2}}$**... (i)**

Electric field at P is

$E = -\frac{dv}{dx} = 1.8 \times 10^4 8 \frac{-1}{2} \frac{27}{2} + x^{2\,-\frac{3}{2}} - -\frac{1}{2} \frac{3}{2} + x^{2\,-\frac{3}{2}} 2x$

E = 0 on x -axis where x = 0 or

$\frac{8}{\frac{27}{2} + x^{2\,\frac{3}{2}}} = \frac{1}{\frac{3}{2} + x^{2\,\frac{3}{2}}}$

$\Rightarrow \frac{4^{\frac{3}{2}}}{\frac{27}{2} + x^{2\,\frac{3}{2}}} = \frac{1}{\frac{3}{2} + x^{2\,\frac{3}{2}}}$

$\Rightarrow \frac{27}{2} + x^2 = 4\frac{3}{2} + x^2$

This equation gives $x = \pm \sqrt{\frac{5}{2}} m$

The least value of kinetic energy of the particle at infinity should be enough to take the particle upto x = $+\sqrt{\frac{5}{2}} m$

because at x = $+\sqrt{\frac{5}{2}} m$**, E = 0.**

=> Electrostatic force on charge q is zero or $F_e = 0$.

For at x > $\sqrt{\frac{5}{2}} m$**, E is repulsive (towards positive x - axis)**

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and for x < $\sqrt{\frac{5}{2}}$m, E is attractive (towards negative x - axis)

Now, from Eq. (i), potential at x = $\sqrt{\frac{5}{2}}$m

V = 1. 8x 10^4 $\frac{8}{\sqrt{\frac{27}{2}+\frac{5}{2}}} - \frac{1}{\sqrt{\frac{3}{2}+\frac{5}{2}}}$

V = 2. 7 x 10^4 volt

Applying energy conservation at x = ∞ and x = $\sqrt{\frac{5}{2}}$m

$\frac{1}{2}$ **$mv^2_0 = q_0Vp$ ... (ii)**

v_0 = $\sqrt{\frac{2q_0V}{m}}$

Substituting the values v_0 = $\sqrt{\frac{2\times10^{-7}\times2.7\times10^4}{6\times10^{-4}}}$

v_0 = 3 m/s Therefore Minimum value of v_0 is 3 m/s

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